

Chalo aapki input string `s = "() [ () { } ]"` par is pure code ka **step-by-step dry run** dekhte hain taki aapko samajh aaye ki stack memory backend mein kaise scale aur shrink ho rahi hai.

🧠 Initialization Stage

- String `s = "() [ () { } ]"`
- String Length `n = 8`
- Empty Stack `st = []`

🔍 Index-by-Index Dry Run

Index 0: `i = 0`, `s[0] = '('`

- Condition Check:** Yeh ek opening bracket hai → `if (s[i] == '(' || ...)` satisfies.
- Operation:** `st.push('(')`
- Stack State:** `['(']` (Top par (hai)

Index 1: `i = 1`, `s[1] = ')'`

- Condition Check:** Closing bracket hai, toh `else` block chalega.
- Safety Check:** `st.empty()` is false (safe to proceed).
- Operation:** * `ch = st.top()` → `ch` ban gaya `'('`
 - `st.pop()` → Stack se remove ho gaya.
 - Check match: `(s[1] == ')' && ch == '(')` → **Valid Match!** Loop continues.
- Stack State:** `[]` (Stack khali ho gaya)

Index 2: `i = 2`, `s[2] = '['`

- Condition Check:** Opening bracket.
- Operation:** `st.push('[')`
- Stack State:** `['[']`

Index 3: `i = 3`, `s[3] = '('`

- Condition Check:** Opening bracket.
- Operation:** `st.push('(')`
- Stack State:** `['[', '(']` (Top par ab (hai)

Index 4: `i = 4` , `s[4] = ')'`

- **Condition Check:** Closing bracket → `else` block.
- **Operation:**
 - `ch = st.top()` → `ch` ban gaya `'('`
 - `st.pop()`
 - Check match: `(s[4] == ')' && ch == '(')` → **Valid Match!**
- **Stack State:** `['[']` (Puraana `[` abhi bhi top par bacha hai)

Index 5: `i = 5` , `s[5] = 'ξ'`

- **Condition Check:** Opening bracket.
- **Operation:** `st.push('ξ')`
- **Stack State:** `['[', 'ξ']` (Top par `ξ` hai)

Index 6: `i = 6` , `s[6] = 'ξ'`

- **Condition Check:** Closing bracket → `else` block.
- **Operation:**
 - `ch = st.top()` → `ch` ban gaya `'ξ'`
 - `st.pop()`
 - Check match: `(s[6] == 'ξ' && ch == 'ξ')` → **Valid Match!**
- **Stack State:** `['[']`

Index 7: `i = 7` , `s[7] = ']'`

- **Condition Check:** Closing bracket → `else` block.
- **Operation:**
 - `ch = st.top()` → `ch` ban gaya `'['`
 - `st.pop()`
 - Check match: `(s[7] == ']' && ch == '[')` → **Valid Match!**
- **Stack State:** `[]`